

Two Routes from Additive to Relative Accuracy in Gaussian Boson Sampling

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Gaussian boson sampling complexity arguments connect finite interferometers to Gaussian matrix ensembles while controlling output probabilities. At a common absolute threshold, we compare two routes from additive to relative accuracy. One uses a finite transpose Gram reference, and the other an independent complex symmetric Gaussian reference. We derive relative error failure bounds separating additive estimation, anticoncentration, and matrix law replacement errors, and identify a regime in which the finite transpose Gram route yields a vanishing relative error certificate.

Introduction. Quantum random sampling links experimentally accessible probability distributions to counting problems believed to be classically intractable [1, 2]. Gaussian boson sampling (GBS) realizes this program using squeezed Gaussian states and photon counting; collision free probabilities contain squared hafnians [3, 4]. Quantum advantage arguments separate an ensemble replacement step, a reduction from sampling to estimation, and an average case hardness input. Relative estimation reductions generally invoke anticoncentration explicitly. Some GBS proposals instead formulate the remaining average case assumption directly as an additive error hafnian estimation conjecture [5, 6]. The companion Articles supply two routes. The first retains a finite dimensional Gaussian transpose Gram reference, while the second uses an independent complex symmetric Gaussian reference [7, 8]. This Letter compares their bounds at a common absolute threshold, identifies the parameter regime in which their guarantees differ, and states the additional complexity assumptions needed for sampling consequences.

Physical setting and established bounds. Let U be Haar distributed in $U(M)$. Prepare K identical single mode squeezed vacuum inputs with squeezing strength $s > 0$, leaving the other inputs in vacuum. Fix a collision free pattern S with one photon in each of $N = 2n$ selected output modes, and let $U_{S,K} \in \mathbb{C}^{N \times K}$ be the corresponding block. Put $A = U_{S,K} U_{S,K}^\top$. The ideal pattern probability is [3, 4] [6, Eq. (1.1)]

$$p_S(U; s) = \frac{\tanh(s)^N}{\cosh(s)^K} |\text{haf}(A)|^2. \quad (1)$$

Here and throughout, $^\top$ denotes ordinary transpose, while a superscript asterisk denotes conjugate transpose. This is a sector with fixed photon number N . Whether that sector is typical depends on s ; the mean number of injected photons is $K \sinh^2 s$. We first prove the bound for one fixed pattern and then average it over the entire collision free output pattern space. That uniform combinatorial average must not be confused with drawing an output from the probability distribution $p_S(U; s)$.

Route 1: finite K transpose Gram. Its hiding ingredient is the uniform product matrix theorem. Write $C_* := 615172$ for its universal constant. If $G \in \mathbb{C}^{N \times K}$

is standard circular complex Gaussian, arbitrary S is allowed: a fixed output row permutation sends S to the first N positions, while Haar left invariance and coordinate permutation invariance of both Gaussian laws reduce the comparisons to the canonical top block. For every $1 \leq N \leq K \leq M$, where throughout $d_{\text{TV}}(\mu, \nu) := \sup_E |\mu(E) - \nu(E)|$,

$$\begin{aligned} d_{\text{TV}}(\text{Law}(MU_{S,K} U_{S,K}^\top), \text{Law}(GG^\top)) \\ \leq \min \left\{ 1, C_* \frac{N^2}{M} \right\}. \end{aligned} \quad (2)$$

This is Theorem I.1 of Ref. [7].

Its anticoncentration ingredient is intrinsic to the Gaussian Gram ensemble. Let

$$\begin{aligned} H_{K,n} &= \text{haf}(GG^\top), \\ \sigma_{K,n}^2 &= \mathbb{E}|H_{K,n}|^2 = (2n-1)!! \prod_{r=0}^{n-1} (K+2r). \end{aligned} \quad (3)$$

Equation (3) was first proved by Ehrenberg et al. [9, Theorem 1 and Supplemental Sec. S3]. Because $N = 2n$, the condition $K \geq 4n$ is exactly $K \geq 2N$. We retain the $4n$ form when quoting the companion anticoncentration theorem and use $2N$ in the comparison in terms of N . For $n \geq 1$, $K \geq 4n$, every $z \in \mathbb{C}$, and $\varepsilon \geq 0$,

$$\mathbb{P}\{|H_{K,n} - z| \leq \varepsilon \sigma_{K,n}\} \leq \min\{1, B_{K,n} \varepsilon^2\}, \quad (4)$$

where

$$\begin{aligned} B_{K,n} &= b_n \frac{K}{K-1} \prod_{r=2}^n \frac{K+2r-2}{K-4r+1}, \\ b_n &= \frac{2\Gamma(n+1/2)}{\sqrt{\pi}\Gamma(n)}. \end{aligned} \quad (5)$$

Equations (4) and (5) are Theorem I.1 of Ref. [8]. The exact product further gives

$$B_{K,n} \leq 4\sqrt{n} \exp\left(\frac{32n^2}{K}\right), \quad n \geq 1, \quad K \geq 4n. \quad (6)$$

This is Eq. (C.3) of Ref. [8]. Reference [10] identified the same $K \asymp n^2 / \log n$ scaling order boundary for weak anticoncentration of the independent Gaussian Gram

hafnian surrogate, defined by an inverse polynomial moment ratio rather than a high probability small ball bound.

Route 2: independent symmetric Gaussian. Its target is the complex symmetric Gaussian matrix G_N^{sym} : its entries on and above the diagonal are independent, its nondiagonal entries are standard circular complex Gaussian, and its diagonal law is defined by multiplying independent standard circular coordinates by square root two. For the fixed block with N rows $U_{S,K}$, Theorem 1.1 of Shou et al. [6] proves, as $K \rightarrow \infty$ with $N \leq K \leq M$,

$$d_{\text{TV}}\left(\text{Law}\left(\frac{M}{\sqrt{K}}A\right), \text{Law}(G_N^{\text{sym}})\right) \leq C' \frac{N}{\sqrt{K}}. \quad (7)$$

Here $C' > 0$ denotes the constant implicit in the cited $O(N/\sqrt{K})$ estimate. All later uses of Eq. (7) are restricted to the same asymptotic regime. That equation is taken from Ref. [6]. Reference [11] proves its scalar consequences under that assumption.

For $N = 2n$, put $H_n^{\text{sym}} = \text{haf}(G_N^{\text{sym}})$ and $(\sigma_n^{\text{sym}})^2 = (2n-1)!!$. For all $z \in \mathbb{C}$ and $\varepsilon \geq 0$,

$$\mathbb{P}\{|H_n^{\text{sym}} - z| \leq \varepsilon \sigma_n^{\text{sym}}\} \leq \min\{1, b_n \varepsilon^2\}. \quad (8)$$

This is Theorem I.3 of Ref. [8]; its proof is in the first subsection of Appendix D of that Article. The universal coefficient bound is $b_n \leq 2\sqrt{n/\pi}$.

The following finite Haar compositions supply the relative error certificates in Fig. 1 and Eq. (18).

Route 1: finite K transpose Gram composition. Let $n \geq 1$, $N = 2n$, and $4n \leq K \leq M$. For every $z \in \mathbb{C}$ and $\varepsilon \geq 0$,

$$\begin{aligned} \mathbb{P}_U\{|\text{haf}(MU_{S,K}U_{S,K}^\top) - z| \leq \varepsilon \sigma_{K,n}\} \\ \leq \min\left\{1, B_{K,n}\varepsilon^2 + C_* \frac{N^2}{M}\right\}. \end{aligned} \quad (9)$$

Proof. See End Matter, “Two single pattern compositions and their probability scales,” Eq. (28).

Route 2: independent symmetric Gaussian composition. For $n \geq 1$, $N = 2n$, $N \leq K \leq M$, and sufficiently large K in the regime of Eq. (7), every $z \in \mathbb{C}$ and $\varepsilon \geq 0$ satisfy

$$\begin{aligned} \mathbb{P}_U\left\{\left|\text{haf}\left(\frac{M}{\sqrt{K}}U_{S,K}U_{S,K}^\top\right) - z\right| \leq \varepsilon \sigma_n^{\text{sym}}\right\} \\ \leq \min\left\{1, b_n \varepsilon^2 + C' \frac{N}{\sqrt{K}}\right\}. \end{aligned} \quad (10)$$

Proof. The same End Matter argument, Eq. (28), applies with Eq. (7) as the cited matrix law input.

We next translate both amplitude theorems into probability statements. The Route 1 Gaussian reference scale is

$$p_{\text{ref}}^{(1)} := p_{\text{ref}} := \frac{\tanh(s)^N}{M^N \cosh(s)^K} \sigma_{K,n}^2. \quad (11)$$

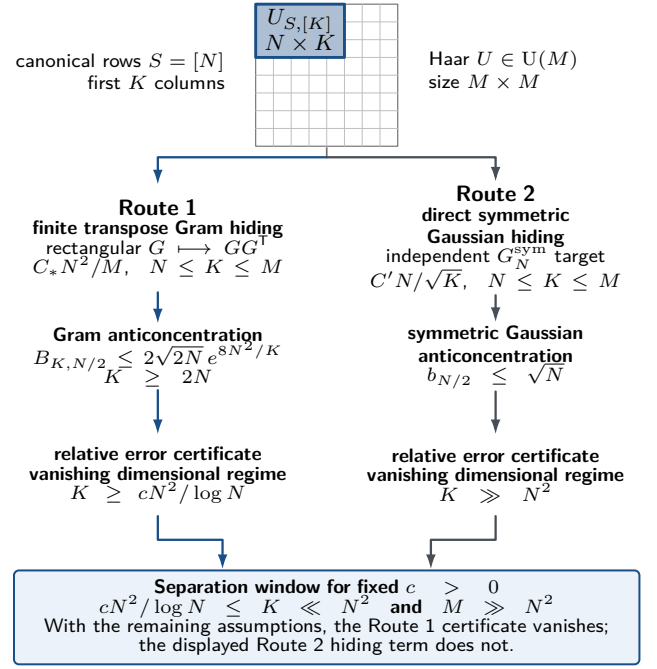


FIG. 1. Two pipelines from the same $N \times K$ Haar block to relative error. Route 1 forms the transpose Gram target $GG^\top$ from a rectangular Gaussian matrix, whereas Route 2 uses a direct independent symmetric Gaussian target. The graphic isolates the dimensional conditions; the remaining assumptions are stated in the text. Both use the same absolute threshold but have different reference probabilities. The lower strip shows the separation window and the shared collision free scaling $M \gg N^2$.

Because the hafnian of an $N \times N$ matrix is homogeneous of degree $N/2 = n$, Eqs. (1) and (11) imply $p_S/p_{\text{ref}} = |\text{haf}(MA)|^2/\sigma_{K,n}^2$. Taking $z = 0$ and $\varepsilon = \sqrt{t}$ in Eq. (9) gives, for every $t \geq 0$,

$$\mathbb{P}_U\{p_S \leq tp_{\text{ref}}\} \leq \min\left\{1, B_{K,n}t + C_* \frac{N^2}{M}\right\}. \quad (12)$$

Route 2 has the natural reference scale

$$p_{\text{ref}}^{(2)} := \frac{\tanh(s)^N}{\cosh(s)^K} \frac{K^n}{M^N} (\sigma_n^{\text{sym}})^2. \quad (13)$$

The two scales are not identical. Their exact ratio is

$$p_{\text{ref}}^{(1)} = R_{K,n} p_{\text{ref}}^{(2)}, \quad R_{K,n} := \prod_{r=0}^{n-1} \left(1 + \frac{2r}{K}\right). \quad (14)$$

Hafnian homogeneity and Eq. (10) give

$$\mathbb{P}_U\{p_S \leq tp_{\text{ref}}^{(2)}\} \leq \min\left\{1, b_n t + C' \frac{N}{\sqrt{K}}\right\}. \quad (15)$$

Fair comparison from additive to relative error. Let $\tilde{p}(U, \mathbf{S}, \omega)$ be one fixed measurable randomized estimate, where U is Haar and $\mathbf{S}$ is independently uniform over the

collision free patterns of size N . Haar row exchangeability makes each ideal small denominator tail above identical for every fixed pattern, and therefore preserves it after averaging over the independent uniform label. The additive error tail below is defined jointly for the chosen estimator and is not inferred from exchangeability. The internal randomness ω may depend on $(U, \mathbf{S})$. Write $\Delta p = \tilde{p} - p_{\mathbf{S}}$, fix $\rho > 0$, and fix one common absolute threshold $\tau \geq 0$. Thus both routes use the same tail $\gamma(\tau) := \mathbb{P}\{|\Delta p| > \tau\}$ and the same deterministic inclusion

$$\{|\Delta p| > \rho p_{\mathbf{S}}\} \subseteq \{|\Delta p| > \tau\} \cup \{p_{\mathbf{S}} \leq \tau/\rho\}. \quad (16)$$

To avoid ambiguity, set precisely $p_1 := p_{\text{ref}}^{(1)}$ and $p_2 := p_{\text{ref}}^{(2)}$; these abbreviations are used throughout the comparison. In the common range $4n \leq K \leq M$, put $F_\rho := \mathbb{P}\{|\Delta p| > \rho p_{\mathbf{S}}\}$. The resulting directly comparable bound is

$$\begin{aligned} F_\rho &\leq E_{\text{fair}}(\tau) := \min\{E_1(\tau), E_2(\tau)\}, \\ E_1(\tau) &:= \min\left\{1, \gamma(\tau) + B_{K,n} \frac{\tau}{\rho p_1} + C_* \frac{N^2}{M}\right\}, \\ E_2(\tau) &:= \min\left\{1, \gamma(\tau) + b_n \frac{\tau}{\rho p_2} + C' \frac{N}{\sqrt{K}}\right\} \\ &= \min\left\{1, \gamma(\tau) + b_n R_{K,n} \frac{\tau}{\rho p_1} + C' \frac{N}{\sqrt{K}}\right\}. \end{aligned} \quad (17)$$

The last identity uses $p_1 = p_{\text{ref}}^{(1)} = R_{K,n} p_{\text{ref}}^{(2)} = R_{K,n} p_2$. It shows exactly why imposing equal values of η_1/ρ and η_2/ρ would not be fair: it would compare the different absolute thresholds $\eta_1 p_1$ and $\eta_2 p_2$, and hence generally different additive tails. At the same τ , instead, $\tau/(\rho p_2) = R_{K,n} \tau/(\rho p_1)$; equivalently, the same physical threshold requires $\eta_2 = R_{K,n} \eta_1$.

To make the precision dependence explicit, put $\eta := \tau/p_1$ and impose the common polynomial scale $\eta/\rho = N^{-a}$ for a fixed $a > 0$. Equation (17) then becomes

$$\begin{aligned} E_1(\tau) &= \min\left\{1, \gamma(\tau) + B_{K,n} N^{-a} + C_* \frac{N^2}{M}\right\}, \\ E_2(\tau) &= \min\left\{1, \gamma(\tau) + b_n R_{K,n} N^{-a} + C' \frac{N}{\sqrt{K}}\right\}. \end{aligned} \quad (18)$$

Here $N = 2n$; the pair count n is retained only inside the exact companion coefficients $B_{K,n}$, b_n , and $R_{K,n}$.

At the same absolute threshold, the small denominator coefficients obey the exact comparison

$$\frac{B_{K,n}}{b_n R_{K,n}} = \frac{K}{K-1} \prod_{r=2}^n \frac{K}{K-4r+1} > 1, \quad K \geq 4n. \quad (19)$$

Thus Route 2 has the smaller anticoncentration coefficient; the advantage identified below for Route 1 comes instead from its hiding requirement.

For the asymptotic statements, consider instances indexed by n : set $N = 2n \rightarrow \infty$, write $K = K_n$,

$M = M_n$, denote the corresponding $p_1, \Delta p$ by $p_{1,n}, \Delta p_n$, and choose $\tau = \tau_n$ so that $\tau_n/(\rho p_{1,n}) = N^{-a}$. Write $\gamma_n := \mathbb{P}\{|\Delta p_n| > \tau_n\}$, and assume $4n \leq K_n$ eventually. If $\kappa > 0$ is fixed and $N^2/K_n \leq \kappa \log N$ eventually, then

$$\begin{aligned} B_{K_n,n} &= O\left(N^{1/2+3\kappa/4}\right), \\ R_{K_n,n} &= O\left(N^{\kappa/4}\right), \\ b_n R_{K_n,n} &= O\left(N^{1/2+\kappa/4}\right). \end{aligned} \quad (20)$$

Here and below $O(\cdot)$ is along $n \rightarrow \infty$. In the original pair count notation, $n^2/K_n \leq C \log n$ with fixed $C > 0$ implies $N^2/K_n \leq 4C \log N$. Substituting $\kappa = 4C$ in Eq. (20) gives

$$B_{K_n,n} = O\left(N^{1/2+3C}\right), \quad b_n R_{K_n,n} = O\left(N^{1/2+C}\right). \quad (21)$$

Consequently the Route 1 upper bound tends to zero when $\gamma_n \rightarrow 0$, $N^2/M_n \rightarrow 0$, and $a > 1/2 + 3\kappa/4$. For Route 2, the published hiding envelope is guaranteed to vanish when $N^2/K_n \rightarrow 0$. In that regime,

$$\frac{N^2}{K_n} \rightarrow 0 \implies R_{K_n,n} \rightarrow 1, \quad (22)$$

Thus, for any fixed $c > 0$ and $a > 1/2 + 3/(4c)$, consider the simultaneous conditions

$$c \frac{N^2}{\log N} \leq K_n \ll N^2 \quad \text{and} \quad M_n \gg N^2, \quad (23)$$

where $K_n \ll N^2$ means $K_n/N^2 \rightarrow 0$ and $M_n \gg N^2$ means $N^2/M_n \rightarrow 0$. In this window the displayed Route 1 upper bound tends to zero when $\gamma_n \rightarrow 0$. By contrast, because $C' > 0$ and $K_n/N^2 \rightarrow 0$, the uncapped contribution $C'N/\sqrt{K_n} = C'/\sqrt{K_n/N^2}$ diverges, so the published Route 2 upper bound does not tend to zero. This comparison concerns these upper bounds only and gives no lower bound on the actual Route 2 total variation distance. Both the K_n window and the M_n condition are required simultaneously. The condition $M_n \gg N^2$ is also the bosonic birthday scale for standard boson sampling with one photon in each of N input modes: Aaronson and Arkhipov bound the Haar average collision probability by $O(N^2/M_n) = o(1)$ [1, Appendix C]. Collision free dominance for the unconditioned squeezed input GBS distribution additionally requires control of its photon number distribution [5]; that issue is separate from the fixed sector bounds here.

Uniform label transfer from additive to relative error using total variation. Let $[M] = \{1, \dots, M\}$ and let

$$\begin{aligned} \Omega_{M,N} &:= \{S \subseteq [M] : |S| = N\}, \\ D_{M,N} &:= |\Omega_{M,N}| = \binom{M}{N}. \end{aligned} \quad (24)$$

For a fixed U , let P_U be the ideal distribution and Q_U an approximate distribution on the full discrete count pattern space, with probability masses p_x and q_x . Under the convention $d_{TV}(P, Q) = \sup_E |P(E) - Q(E)|$, the TV guarantee controls the mass difference of every finite event. Splitting the finite collision free label set according to the sign of $p_S - q_S$ then gives the factor two below. If $\varepsilon_{\text{samp}} > 0$, $d_{TV}(P_U, Q_U) \leq \varepsilon_{\text{samp}}$, and $\mathbf{S} \sim \text{Unif}(\Omega_{M,N})$, then the standard approximate sampling step of Aaronson and Arkhipov [1, Eqs. (5.81) through (5.87)] gives the uniform pattern mean absolute probability error

$$\begin{aligned} \mathbb{E}_{\mathbf{S}} |p_{\mathbf{S}} - q_{\mathbf{S}}| &= \frac{1}{D_{M,N}} \sum_{S \in \Omega_{M,N}} |p_S - q_S| \\ &\leq \frac{2d_{TV}(P_U, Q_U)}{D_{M,N}} \leq \frac{2\varepsilon_{\text{samp}}}{D_{M,N}}. \end{aligned} \quad (25)$$

Markov's inequality therefore gives, for every $0 < \zeta < 1$,

$$\mathbb{P}_{\mathbf{S}} \left\{ |p_{\mathbf{S}} - q_{\mathbf{S}}| > \frac{2\varepsilon_{\text{samp}}}{\zeta D_{M,N}} \right\} \leq \zeta. \quad (26)$$

If the TV guarantee holds for every U , put $F_{\text{samp}} := \mathbb{E}_U \mathbb{P}_{\mathbf{S}} \{ |p_{\mathbf{S}} - q_{\mathbf{S}}| > \rho p_{\mathbf{S}} \mid U \}$. The two routes give the explicit bounds

$$\begin{aligned} F_{\text{samp}} &\leq \min \left\{ 1, \zeta + \frac{2B_{K,n}\varepsilon_{\text{samp}}}{\rho\zeta D_{M,N}p_1} + C_* \frac{N^2}{M} \right\}, \\ F_{\text{samp}} &\leq \min \left\{ 1, \zeta + \frac{2b_n\varepsilon_{\text{samp}}}{\rho\zeta D_{M,N}p_2} + C' \frac{N}{\sqrt{K}} \right\}. \end{aligned} \quad (27)$$

These are probability inequalities, not an efficient estimator. If Q_U is generated by an efficient classical sampler using polynomially many random bits, the approximate counting procedure of Ref. [12], in the form used in Theorem 4.1 and Eqs. (5.89) onward of Ref. [1], can approximate $q_{\mathbf{S}}$ to inverse polynomial multiplicative accuracy. That algorithmic step and the average case hardness assumption remain external to the present bounds.

DATA AND SUPPORTING MATERIAL AVAILABILITY

No experimental data were created. Supporting materials, the complete equation crosswalk, executable audits, and source code are available in Ref. [11]. All theoretical results and labeled equations in this Article have been formalized and verified in Lean, conditional only on the explicitly identified external assumptions.

The formal trust boundary is exact. Every labeled equation and mathematical deduction has a named Lean formalization checked by the kernel. The complete dependency boundary is as follows. Route 1 imports the companion uniform hiding theorem of Ref. [7], which is conditional on A1, A2, A3, and A4, and the finite shifted

transpose Gram anticoncentration theorem of Ref. [8]. Route 2 imports the direct independent symmetric Gaussian small ball theorem from the latter work and uses the Shou et al. [6] matrix comparison, Eq. (7), as one explicit premise. The two anticoncentration theorems use only standard mathematical foundations. Consequently, Eqs. (10), (15), the Route 2 rows of Eqs. (17) and (27), and the Route 2 line of Eq. (28) remain conditional on the Shou premise. Every deduction from the named inputs is checked by the Lean kernel. This certification does not prove the external inputs or independently establish error free correspondence between the prose and formal statements.

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END MATTER: PROOFS OF THE MAIN BOUNDS

This End Matter proves the deductions used in the Letter. The uniform transpose Gram hiding theorem is Theorem I.1 of Ref. [7]. The exact Gram hafnian scale was first proved in Theorem 1 of Ref. [9]. The shifted finite K anticoncentration theorem is Theorem I.1 of Ref. [8], with proof in Sec. V A and Appendix C. The independent symmetric Gaussian small ball bound is Theorem I.3 of Ref. [8], with proof in the first subsection of Appendix D. The comparison between the Haar block and a symmetric Gaussian matrix is Theorem 1.1 of Ref. [6].

Two single pattern compositions and their probability scales. If two matrix laws differ by at most δ in total variation, their pushforwards by every measurable map differ by at most δ . Applying this to the hafnian, put $Y_1 := \text{haf}(MA)$, $Z_1 := \text{haf}(GG^T)$, $Y_2 := \text{haf}(MA/\sqrt{K})$, and $Z_2 := \text{haf}(G_N^{\text{sym}})$. Also put $\mathcal{D}_1 := \{w \in \mathbb{C} : |w - z| \leq \varepsilon \sigma_{K,n}\}$ and $\mathcal{D}_2 := \{w \in \mathbb{C} : |w - z| \leq \varepsilon \sigma_n^{\text{sym}}\}$. The two disk transfers are

$$\begin{aligned} \mathbb{P}_U\{Y_1 \in \mathcal{D}_1\} &\leq \mathbb{P}_G\{Z_1 \in \mathcal{D}_1\} + C_* \frac{N^2}{M}, \\ \mathbb{P}_U\{Y_2 \in \mathcal{D}_2\} &\leq \mathbb{P}_{G^{\text{sym}}}\{Z_2 \in \mathcal{D}_2\} + C' \frac{N}{\sqrt{K}}. \end{aligned} \quad (28)$$

The two Gaussian bounds prove Eqs. (9) and (10). At $z = 0$, hafnian homogeneity converts the radius into the intensity thresholds tp_1 and tp_2 with $t = \varepsilon^2$, giving Eqs. (12) and (15). For Route 2, the second line uses the matrix law estimate in Eq. (7), taken from Ref. [6].

The reference scale identity follows by dividing the exact finite K variance by $K^n(\sigma_n^{\text{sym}})^2$:

$$\frac{\sigma_{K,n}^2}{K^n(\sigma_n^{\text{sym}})^2} = \prod_{r=0}^{n-1} \left(1 + \frac{2r}{K}\right) = R_{K,n}. \quad (29)$$

This proves Eq. (14). Dividing the exact product for $B_{K,n}$ by $b_n R_{K,n}$ cancels every numerator factor and leaves

$$\frac{K}{K-1} \prod_{r=2}^n \frac{K}{K-4r+1}. \quad (30)$$

Every factor is greater than one when $K \geq 4n$, proving Eq. (19).

For any $\tau \geq 0$, a relative error failure is contained in the union of $\{|\Delta p| > \tau\}$ and $\{p_S \leq \tau/\rho\}$. Applying the two explicit single pattern tails at that same τ , and using $p_1 = R_{K,n}p_2$, proves Eq. (17). Their minimum is $E_{\text{fair}}(\tau)$.

Coefficient asymptotics. For $n \geq 1$ and $K \geq 4n$, apply $\log(1+x) \leq x$ to every factor in the exact products. Direct summation gives

$$\begin{aligned} \log R_{K,n} &\leq \frac{n(n-1)}{K}, \\ \log \frac{B_{K,n}}{b_n} &\leq \frac{3n^2-2}{K} + \frac{9n^3}{K(K-4n+1)}. \end{aligned} \quad (31)$$

For the second line, expand each reciprocal denominator around K^{-1} , sum the first order terms, and use $K-4r+1 \geq K-4n+1$ for the remainder. Under $N = 2n$ and $N^2/K_n \leq \kappa \log N$, eventually $K_n-4n+1 \geq K_n/2$, so the remainder in Eq. (31) is $o(1)$. The first terms give $R_{K,n} = O(N^{\kappa/4})$ and $B_{K,n}/b_n = O(N^{3\kappa/4})$. The Gamma ratio $b_n = O(N^{1/2})$ proves Eq. (20); substituting $\kappa = 4C$ proves Eq. (21). The first line of Eq. (31) tends to zero when $N^2/K_n \rightarrow 0$, and exponentiation proves Eq. (22).

Uniform label transfer from additive to relative error using total variation. The supremum over events TV guarantee applies to every finite subset of the full output space. Splitting $\Omega_{M,N}$ into the labels where $p_S - q_S$ is nonnegative and negative bounds the sum of absolute differences by twice the TV error. Dividing by $D_{M,N}$ proves

$$\mathbb{E}_S |p_S - q_S| \leq \frac{2\varepsilon_{\text{samp}}}{D_{M,N}}. \quad (32)$$

Markov's inequality immediately gives Eq. (26). This is the standard approximate sampling step in Eqs. (5.81) through (5.87) of Aaronson and Arkhipov, cited above; approximate counting is a subsequent ingredient, not part of this inequality. Combining the result with the preceding inclusion involving two events gives Eq. (27).